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Underwater Object Detection using Image Enhancement and Deep Learning Models

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Abstract—Autonomous underwater vehicles (AUVs) are efficient robotic tools, offering a wide range of applications in ocean exploration and research, such as oceanographic mapping, environmental monitoring, and archaeology. Incorporating an automatic object detection system with AUVs can substantially improve their ability to perceive and recognize objects in a complicated and often hazardous environment. Currently, detecting underwater objects relied on a man-in-the-loop approach, where AUVs captured vast amounts of data and saved them in memory for offline processing. This study investigates the use of deep learning for automatic image preprocessing and object detection, evaluating and comparing three state-of-the-art YOLO (You Only Look Once) models, including YOLOv8, YOLOv7, and YOLOv5. Extensive experiments were conducted using publicly available underwater image datasets, revealing that the pre-trained models attain superior performance on the Brackish dataset. YOLOv5 and YOLOv8 achieved the highest mean average precision (mAP) with a score of 99%, while YOLOv7 scored 89%. Furthermore, an underwater image enhancement algorithm is employed on the URPC2021 dataset, significantly improving the detection accuracy with a 3% increase in mAP across all three models. In terms of inference speed, YOLOv5 demonstrated the highest frames per second (FPS), while maintaining comparable performance in mAP and recall.

Keywords—Underwater robotics, AUV, underwater object detection, image enhancement, YOLOv8, YOLOv7, YOLOv5.

I. INTRODUCTION

The marine environment is a diverse and intricate part of the Earth's surface, serving a significant role in sustaining both the environment and human populations. It provides valuable minerals, oil, gas, and other aquatic resources, making it a target for marine exploration endeavours [1]. However, its harsh conditions hinder exploration through traditional means, rendering it the least explored environment. In recent years, the development of underwater robots, such as autonomous underwater vehicles (AUVs) provides a great opportunity to explore and protect the resources beneath the water. AUVs come with various sensing devices, including underwater cameras, sonars, depth sensors, and lighting. They are also equipped with other payload devices that enable them to monitor underwater environments and carry out intricate underwater operations. These operations include capturing marine organisms, creating oceanographic maps, inspecting

pipes and cables, conducting environmental surveillance, and exploring wrecks and archaeological sites. The flourishing growth of artificial intelligence (AI) and intelligent systems are indispensable technologies to accomplish these tasks and play an important role in the development of AUVs. Integrating an intelligent object detection system on board can significantly enhance the perception and recognition capability of AUVs.

Underwater object detection methods rely on either acoustic images or optical images [2]. Sonars and vision cameras are key perception equipment used to identify and detect objects in underwater environments. In contrast to sonars, optical images captured by vision cameras offer higher resolution and a greater amount of detailed information [3]. Moreover, optical systems are more cost-effective in terms of acquisition methods. As a result, there is an increasing interest in using optical systems for underwater target detection.

Traditionally, the task of detecting underwater objects was performed by a man-in-the-loop approach where AUVs capture imaging data and store them in memory for offline processing by expert analysis [4]. However, there is an increasing demand for automatic underwater processing to enable on the fly decision-making and to extend mission times. Specifically, undersea exploration using automatic object detection has two advantages. Firstly, it allows AUVs to make real-time decisions based on the data it collects where accurate detection and recognition of objects undersea is imperative, thereby saving a lot of time and allowing longer surveys. Secondly, real-time underwater object detection can enable greater autonomy for AUVs, which perform preprogrammed missions. AI-powered AUVs are expected to perform not only to collect data but also to perceive and react to the data it collects immediately (e.g., reinspection of interesting objects).

In the last few years, deep learning (DL) techniques have revolutionized the field of computer vision and have fuelled the practical application of underwater object detection. Villon et al. [5] compared the traditional approach (histogram of oriented gradients + support vector machine) with the deep learning method in coral reef fish detection and their experimental analysis showed the superiority of the deep learning method for object detection underwater. In their

work, Wang et al. [6] introduced a deep learning architecture that incorporates convolutional encoding and decoding features to recognize objects underwater. Their proposed framework utilizes a pre-trained convolutional model, AlexNet, which was initially trained for the ImageNet task. The authors transfer the knowledge of the first two layers of the model to facilitate the underwater detection task. Similarly, another study [7] utilized deep convolutional networks, transfer learning, and data augmentation to develop a real-time fish detection and tracking framework from video monitoring systems of AUVs. Recently, Michael et al. [8] created a method for detecting litter in underwater environments using visual deep learning to tackle the issue of plastic debris pollution. The researchers assessed the effectiveness and precision of different deep learning models, such as Faster RCNN, SSD, YOLOv2, and Tiny-YOLO. Faster RCNN was found to have the best performance, although with a weakened inference time. YOLOv2 achieved a good trade-off between speed and accuracy.

This paper aims to explore the use of the latest deep-learning techniques for automatic object detection in AUVs and to determine the most suitable algorithm for deployment using underwater images and videos. AUVs require real-time decision-making, where correct detection and classification of objects underwater is imperative. Detection based on classical computer vision is difficult and error-prone due to manually created feature extractions. The key highlights of our study are outlined in the following.

- An underwater image enhancement pipeline that incorporates colour correction, dehazing, and contrast enhancement is developed to enhance the quality of underwater images and improve detection accuracy.
- Evaluate the performance of three state-of-the-art YOLO models (YOLOv5, YOLOv7, and YOLOv8) for detecting marine objects in challenging underwater conditions.
- Conduct a comprehensive experimental study on three different underwater benchmark datasets to determine the most effective object detection approach for AUVs.

II. METHODS AND MATERIALS

A. Overall Framework

Fig.1 presents the overall framework of the underwater object detection network proposed in this study. Three publicly available underwater image datasets were employed to train the YOLO (You Only Look Once) models [9], providing a diverse range of images with varying lighting conditions, water depths, and underwater scenes. Initially, images undergo pre-processing via augmentation and an underwater image-enhancement pipeline. The resulting image, along with the original image, is then utilized as input data for the object-detection network. Three YOLO frameworks (namely, YOLOv5, YOLOv7, and YOLOv8) are

used to detect and recognize objects. The performance of the trained models was evaluated on the test sets of the three publicly available underwater image datasets.

B. Underwater Datasets

This paper employs three publicly available datasets for underwater object detection. The underwater robot professional contest 2021 (URPC2021) dataset [10], the Brackish dataset [11], and the Aquarium dataset [12] were used for training underwater object detection algorithms. Fig.2 illustrates the statistical data that displays the number of targets in each dataset.

The URPC2021 dataset is an underwater robot professional contest dataset of 2021, which was created to evaluate the performance of underwater object detection algorithms. The dataset consists of 8200 underwater images that were extracted from videos captured by an underwater robot ROV in natural environments. The dataset includes box-level annotations for four categories of objects: holothurian, echinus, starfish, and scallops. The echinus category stands as the most prevalent class, followed by starfish, holothurian, and scallop, in terms of abundance, as shown in Fig.2 (a).

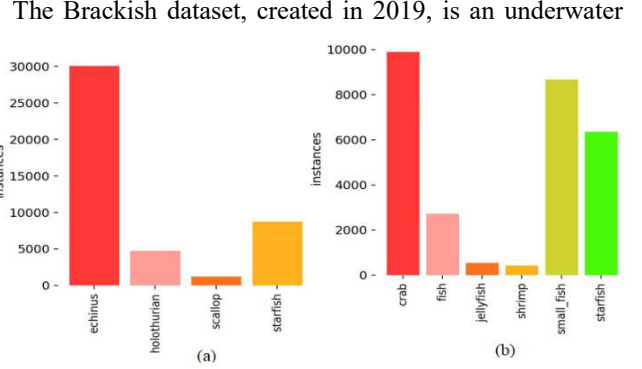


Fig.2. Statistical distribution of targets in each dataset. (a) URPC2021 dataset. and (b) Brackish dataset

dataset comprising more than 14,000 frames. It was created by annotating real filmed underwater videos and encompasses six distinct classes of underwater objects: Big fish, Crab, Jellyfish, Shrimp, Small fish, and Starfish. The dataset was collected using three mounted cameras positioned on the seabed, resulting in a diverse collection of images and viewpoints.

The Aquarium dataset is relatively smaller, consisting of only 638 images collected from two aquariums. However, it still contains multiple bounding boxes with seven different classes of underwater objects, which include fish, jellyfish, penguin, puffin, shark, starfish, and stingray. The dataset was labelled for object detection.

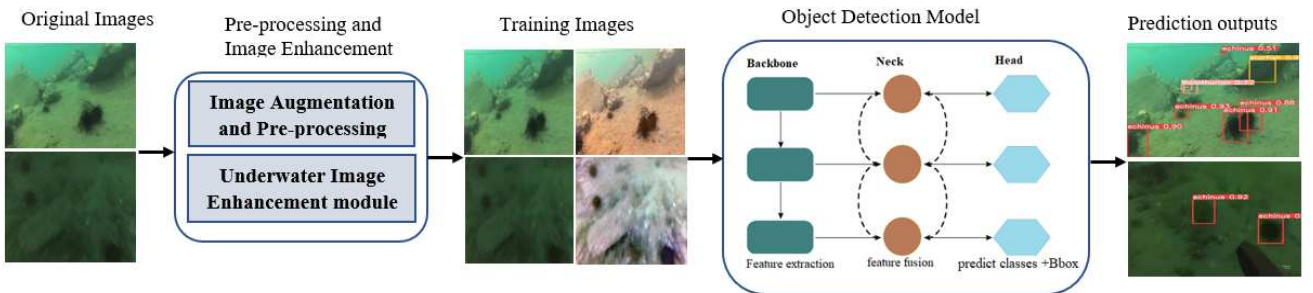


Fig.1. General framework of the object detection architecture

C. Underwater Image Preprocessing

Images captured underwater suffer from low visibility and colour distortions caused by light scattering by particles in the water and wavelength-dependent light absorption, unlike images taken on the surface. Light absorption results in significant colour distortion and loss of image information, while light scattering produces haze effects, suppresses image details, and reduces image contrast [13]. Detecting underwater objects using cameras is challenging due to these negative effects, as well as other complex background interferences such as camera shaking and non-uniform illumination, affecting real-time detection performance underwater. Fig. 3 shows some low-quality underwater images taken from the URPC2021 dataset. Image (a) depicts a low-resolution underwater image. In image (b), a noticeable colour bias is present, and the overall style appears to be dominated by green tones. Image (c) exhibits a haze effect caused by light scattering in underwater environments. The issue with image (d) lies in its low contrast and the presence of a colour cast. To improve the visual quality of such underwater images and to

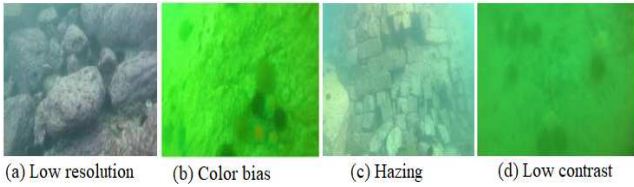


Fig.3 Example of an underwater images on the URPC2021 dataset.

enhance the detection accuracy, underwater image pre-processing, such as image enhancement or restoration, is an essential step [14].

In this study, underwater image enhancement techniques, such as dehazing, colour correction, and contrast enhancement have been applied to remove haze and the colour cast from images [15][16]. The enhancement technique used in this paper is based on a single-image approach that enhances underwater images without requiring prior knowledge of light properties or imaging models [17][18]. The image enhancement module consists of a series of independent processing steps. These steps are designed to effectively correct the degraded images and enhance their quality for improved object recognition.

D. Underwater Object Detectors

The objective of a contemporary object detector is to identify both the location and type of object in every input image. The state-of-the-art detectors consist of three primary components: a backbone that extracts features and generates a feature map representation of the input image through a reliable image classifier, a neck that is connected to the backbone and functions as a feature aggregator by assembling feature maps from various stages of the backbone and integrating these multi-level features, and a head that identifies bounding boxes and conducts classification predictions [19], as shown in Fig. 1.

The present study utilized YOLO architectures for underwater object detection. YOLO has gained widespread use as a real-time object detection system due to its exceptional speed and accuracy, resulting in its popularity in various fields like robotics, autonomous vehicles, and video surveillance. Specifically, three advanced YOLO detectors - YOLOv5, YOLOv7, and YOLOv8 - were compared in this study. YOLOv5 is a single-stage object detection algorithm, which is more efficient and versatile than its earlier iterations [20]. During evaluation on the MS COCO dataset test-dev

2017, YOLOv5 achieved an AP of 50.7% with an image size of 640 pixels. Additionally, YOLOv5 is known for its ease of use, training, and deployment. YOLOv7 is a modified version of YOLOv5, incorporating several enhancements, including the use of residual blocks, skip connections, and anchor boxes, to improve both accuracy and speed while reducing false positives [21]. YOLOv8 [22], which was recently introduced by Ultralytics, claims to be the current leader in real-time object detection. It offers faster processing speeds than previous versions of YOLO and supports state-of-the-art computer vision algorithms, such as instance segmentation and image classification.

E. Model Evaluation Measures

The standard metrics in object detection were used for evaluation and comparison of the models, including, precision, recall, precision-recall curve, average precision (AP), and mean average precision (mAP) with intersection over union (IoU).

Precision is the fraction of correct detections among all the detections made by the model, while Recall is the fraction of correct detections among all the true objects in the scene. Higher values of both metrics indicate better performance.

Intersection over Union (IoU) measures the overlap between the predicted bounding box and the ground truth bounding box. For example, how much of the picture does the predicted bounding box cover? An IoU value of 1.0 indicates a perfect overlap, while values closer to 0 indicate little to no overlap.

Average Precision (AP) measures the average precision across all recall values. A higher AP value indicates better performance. Mean Average Precision (mAP) is the average AP value across all object classes. It is commonly used in object detection competitions to evaluate the overall performance of a model.

- mAP@ 0.5: is the average of AP of all pictures in each category when IoU is set to 0.5.
- mAP@ 0.5:0.95: This is the average of mAP considering different IoU thresholds (from 0.5 to 0.95 in steps of 0.05)

III. EXPERIMENTS AND DISCUSSIONS

This section presents the implementation details and the experimental results of the proposed framework along with detailed discussions.

A. Experimental details

This study utilized the NTNU's IDUN computing cluster [23] for all experiments and implementations. The cluster comprises over 70 nodes and 90 general-purpose graphics processing units (GPUs), each of which is equipped with at least 128 GB of main memory and two Intel Xeon cores and is connected to an Infiniband network. Half of the nodes are fitted with two or more NVIDIA Tesla P100 or V100 GPUs. For training and testing YOLO models, the study employed CUDA 11.7, the PyTorch 2.0 framework, anaconda 3 with Jupyter Notebook, and Python 3.9.12.

The training epochs were set to 150 for the YOLO models. YOLOv5 and YOLOv8 models were trained on input images of size 640×640 . In contrast, YOLOv7 was trained on size of 416×416 input images. All three models utilized stochastic gradient descent (SGD) as the default optimizer. The hyperparameters used for training and testing the models are summarized in Table 1. The image pre-processing method is applied to the training dataset that involves data preparation, noise reduction, augmentation, and image enhancement. In addition, we generated annotations in YOLO format, along

with configuring parameters within the pre-trained models. All three datasets (Aquarium, Brackish and URPC2021) used in this experiment were split into training (80%), validation (10%) and testing (10%).

Table 1. Parameters used in the experiment.

YOLO Model	Batch size	epochs	Learning rate	Weight decay	Input shape
YOLOv5s	32	150	0.01	0.0005	640×640
YOLOv7x	16	150	0.01	0.0005	416×416
YOLOv8	32	150	0.001	0.001	640×640

B. Experimental Results

Before implementing the underwater image enhancement algorithm, we conducted an experiment on the original datasets to identify the dataset that posed a great challenge for our object detection task. Our preliminary analysis of the Aquarium and Brackish datasets revealed that the YOLO models performed exceptionally well, exhibiting high precision, recall, and mean average precision (detailed performance scores can be found in Table 2). Conversely, the models yielded unsatisfactory results on the URPC2021 dataset. The poor results obtained from the models can be attributed to the significant challenges posed by the complex environments present in the URPC2021 dataset. These challenges include low resolution, haze due to motion blur, low contrast and colour cast as well as the frequency of smaller objects in a complex underwater environment. Fig.4 demonstrates an instance of the original distorted images (Fig.4a) from the URPC2021 dataset alongside its corresponding enhanced images (Fig.4b), obtained through the underwater image enhancement algorithm.



Fig. 4. Sample underwater images on the URPC2021 dataset: (a) Original images (b) Enhanced images.

In order to improve the recognition performance of pre-trained YOLO models, a combined approach is employed, utilizing both enhanced images and the original images as inputs for the object-detection network. Fig.5 illustrates the detection results in terms of mAP@0.5, comparing the performance with and without the image enhancement algorithm at 100 epochs. The model trained with image enhancement demonstrates superior performance compared to the model

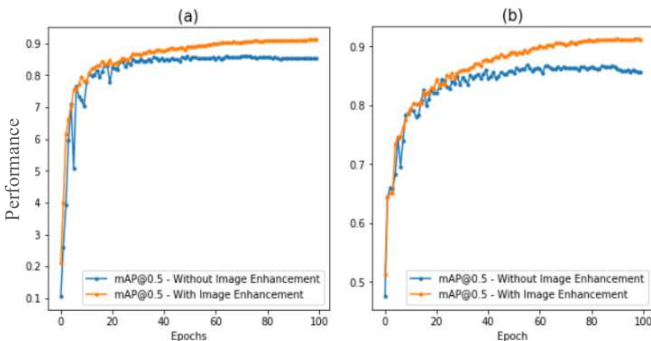


Fig.5. Comparison of mAP@0.5 for 100 epochs. Training with image enhancement shows a greater improvement in mAP compared to training without image enhancement for (a) YOLOv5 and (b) YOLOv8.

trained solely on the raw dataset. Notably, there is a significant increase of 3% in mAP for YOLOv5 and YOLOv7 and a 4% increase for YOLOv8. Fig. 6 shows the visualization of detection results on the URPC2021 dataset, where the different coloured squares in the figure represent the various targets in each YOLO model, such as echinus (red), holothurian (pink), starfish (yellow), and scallop (orange) in YOLOv5 and YOLOv8. YOLOv5 identified 3 bounding boxes for echinus, 2 for starfish, and 1 for holothurian, all with relatively high confidence scores. Despite the demonstrated improvement in performance using the image enhancement algorithm, YOLOv7 and YOLOv8 models still face challenges in accurately detecting Holothurian species, resulting in instances of missed detections. This is evident in Fig.6 where YOLOv5 successfully detects all classes, while YOLOv7 and YOLOv8 failed to detect Holothurian (i.e., missing detection) and exhibit occurrences of false detection in the URPC2021 dataset. The detection results on the brackish are visualized in Fig.7. It was observed that all three models successfully detected all the fish species (shrimp, starfish, and 2 crabs), with YOLOv5 and YOLOv8 having nearly equal recognition performance. Fig.8 illustrates an example of the results of using the testing dataset for the Aquarium dataset and shows the detection output of three different YOLO models, namely YOLOv5, YOLOv7, and YOLOv8. These models were able to detect all fishes and stingrays, demonstrating their flexibility to model and predict on various dimensions and scales. They were also able to handle the complexity of distinguishing between the underwater animals (such as fish and stingray) and the background, a significant challenge in underwater image analysis. Of the three models, YOLOv7 demonstrated the highest confidence value for detecting the stingray, with a value of 0.99. YOLOv8 came in second with a confidence value of 0.94, and YOLOv5 had the lowest confidence of 0.87.

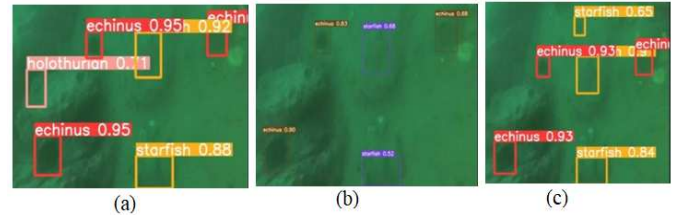


Fig.6. Detection outputs on URPC2021 dataset: (a)YOLOv5, (b) YOLOv7, (c) YOLOv8.

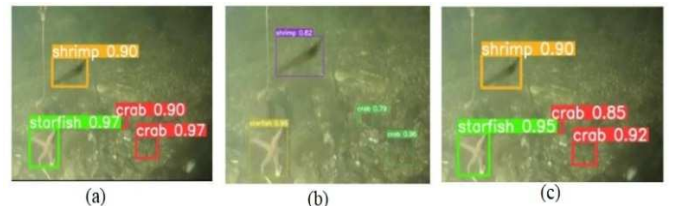


Fig.7. Detection outputs on Brackish dataset: (a)YOLOv5, (b) YOLOv7, (c) YOLOv8.

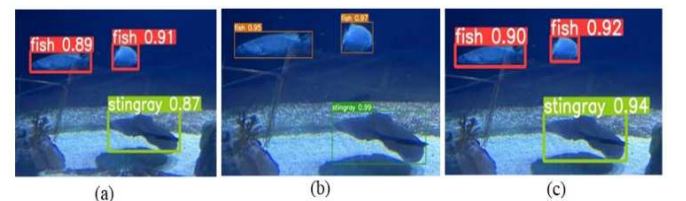


Fig.8. Detection outputs on Aquarium dataset: (a)YOLOv5, (b) YOLOv7, (c) YOLOv8.

The PR curves of all classes for the YOLO models, along with the overall class curve, are depicted in Fig.9 to demonstrate their performance on the Brackish dataset. The average of all mAP classes was used to calculate the overall class curve. YOLOv5 and YOLOv8 exhibited nearly equal performance with the largest area under the curve in the figure, indicating better detection results for all target classes, especially for 'crabs', and 'starfish', with AP values of 99.5%.

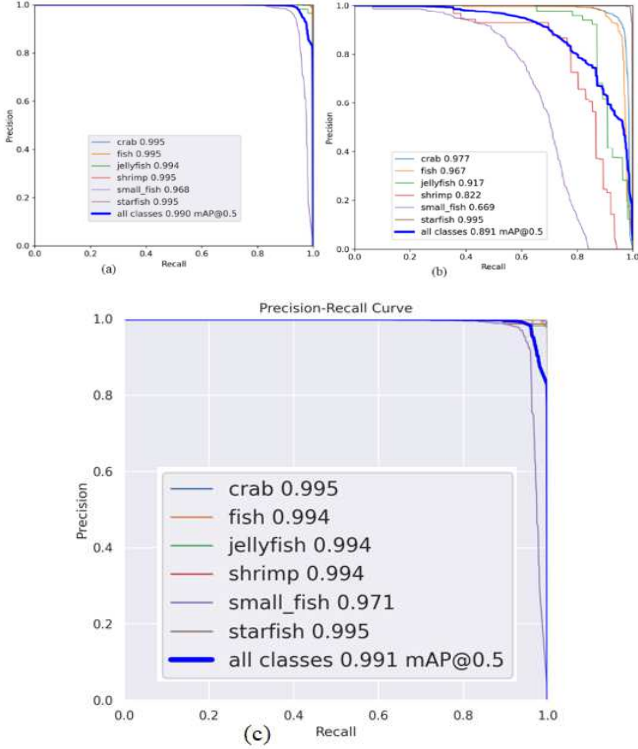


Fig.9. The precision-recall curve of (a)YOLOv5, (b)YOLOv7, and (c)YOLOv8 on Brackish dataset.

The mAP@0.5 value for YOLOv5 and YOLOv8 was 99% approximately. Conversely, YOLOv7 had the lowest area under the PR curve, resulting in a lower mAP value. Certain classes, like 'crab' and 'starfish,' are generally easier to detect due to their prevalence on the seafloor. In contrast, the 'small fish' and 'jellyfish' classes pose greater difficulty for models to learn because they can appear anywhere in the image with similar frequency and have relatively small sizes. Table 2 provides a detailed comparison of the three YOLO models on the original and enhanced dataset of URPC2021, as well as the Brackish and Aquarium datasets. The comparison encompasses various metrics such as accuracy, speed, and latency, providing a detailed analysis of each model's performance across these all the datasets. A closer look at the table, it is evident that the detection results on the URPC2021

enhanced dataset exhibit higher performance in terms of all the accuracy metrics compared to those on the original dataset. This highlights the effectiveness of image enhancement in improving the detection accuracy on the URPC2021 dataset.

On the URPC2021 enhanced dataset and Aquarium dataset, YOLOv7 achieved the highest performance values across all metrics measured. However, when applied to the Brackish dataset, YOLOv7 exhibited lower performance compared to other models. YOLOv5 and YOLOv8 exhibited excellent recognition performance on the Brackish dataset, outperforming YOLOv7 with the highest mean average precision (mAP@0.5) value of 99%. When considering recall, the YOLOv5 model exhibited superior performance compared to YOLOv7 and YOLOv8, achieving a value of 98.5%. On the other hand, YOLOv8 attained the highest performance with mAP@0.5:0.95 value of 85.6%, as shown in Table 2. Although YOLOv8 is claimed to be state-of-the-art and is expected to surpass previous versions of YOLO models in terms of performance, the detection outputs achieved on the three underwater datasets were slightly similar to those obtained with YOLOv5 across all evaluation metrics. However, since the YOLOv8 research is still in progress, it is challenging to exploit its full potential.

In addition to the detection accuracy metrics, the computational complexity of the YOLO models is provided in terms of FPS (frames per second), GFLOPS (giga floating point operations), and parameter size, as shown in Table 2. FLOPs and FPS are metrics that respectively gauge the computational complexity and detection speed of a detector, while parameter size determines the deployability of the detector. In terms of GFLOPS, the YOLOv5 architecture demonstrates an estimated computational cost of 16 GFLOPS, outperforming YOLOv7 and YOLOv8 across all three datasets (Aquarium, Brackish, and URPC2021). This indicates that YOLOv5 requires less computational power for object detection tasks compared to the latest versions. Additionally, YOLOv5 excels in terms of parameter size, making it more suitable for real-time detection and deployable on computing-constrained underwater vehicles such as AUVs (autonomous underwater vehicles). With FPS, the YOLOv5 model achieved the highest FPS on the Aquarium dataset at 135 and on the URPC2021 dataset at 169 when executed on a Tesla V100 GPU. In contrast, YOLOv7 had the lowest FPS on these two datasets. On the Brackish dataset, YOLOv8 outperformed the other YOLO models with the highest execution speed of 162 FPS. However, it had relatively a slower FPS on the URPC2021 dataset.

In general, YOLOv5 can be optimized to deliver competitive detection performance while utilizing fewer FLOPS and achieving higher speeds. This can make it a good choice for AUVs that possess limited computing capability

Table 2: Performance results of the different YOLO models on the different datasets, including the enhanced URPC dataset.

Dataset	Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95	GFLOPS	FPS	Parameters(M)
URPC Original Dataset	YOLOv5	0.869	0.780	0.854	0.664	15.8	169	7.021
	YOLOv7	0.737	0.660	0.724	0.486	188.5	122	70.835
	YOLOv8	0.863	0.784	0.857	0.685	28.4	130	11.138
URPC Enhanced Dataset	YOLOv5	0.903	0.841	0.911	0.700	15.8	169	7.021
	YOLOv7	0.919	0.874	0.923	0.753	188.5	122	70.835
	YOLOv8	0.890	0.851	0.912	0.735	28.4	130	11.127
Aquarium Dataset	YOLOv5	0.968	0.952	0.960	0.754	15.8	135	7.029
	YOLOv7	0.982	0.961	0.966	0.867	189.0	100	70.856
	YOLOv8	0.970	0.953	0.963	0.832	28.5	108	11.128
Brackish Dataset	YOLOv5	0.984	0.985	0.990	0.813	15.8	141	7.029
	YOLOv7	0.911	0.851	0.891	0.606	189.0	103	70.849
	YOLOv8	0.985	0.982	0.990	0.856	28.4	162	11.128

and memory, as it meets their requirements effectively. Despite their slightly lower inference speed compared to YOLOv5, both YOLOv7 and YOLOv8 can provide preferable options for AUVs in terms of robustness and detection performance in complex underwater environments.

IV. CONCLUSIONS

Achieving efficient recognition of objects underwater has been one of the main objectives of autonomous underwater vehicles (AUVs). This paper explores the use of vision-based deep learning algorithms for automatic object detection in (AUVs) using challenging underwater scenes. Three publicly available underwater datasets, Aquarium, Brackish, and UPRC2021, were used to compare the performance of three detection algorithms, namely YOLOv5, YOLOv7, and YOLOv8. An underwater image enhancement pipeline was developed to improve and support the object detection task using these algorithms. The objective was to select the best algorithm or model that could be integrated as a target detection and recognition component of the AUV. The study demonstrates that YOLOv7 achieved superior accuracy and speed in detecting underwater objects on the Aquarium and enhanced version of the UPRC2021 dataset, achieving precision rates of 98% and 92% respectively. However, it struggled with inference time when tested on the Tesla V100 GPU, resulting in slower execution speed compared to other YOLO models. YOLOv8 shows a good balance of accuracy and speed, while YOLOv5 provides the best inference times on GPU. On the Aquarium and UPRC2021 datasets, YOLOv8 and YOLOv5 achieved nearly equal performance in terms of precision, recall, mAP@0.5, and mAP@0.5:0.95, but YOLOv5 was the fastest algorithm that outperformed both YOLOv7 and YOLOv8 across all three datasets.

Overall, the study highlights the potential of vision-based deep learning algorithms in underwater object detection and uses an image enhancement algorithm for improving system performance. The lack of high-quality underwater datasets and images remains a significant challenge in the development of target detection in underwater environments. Future research efforts will focus on optimizing the most effective models by collecting a large and diverse set of underwater datasets and employing image enhancement techniques to improve the overall quality of underwater images, which are crucial for the practicality of the system in real-world applications.

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